Reasoning test

For admission to Class 8

About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 7. Chapters are from the NCERT (CBSE) books a student joining Class 8 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Ganita Prakash

- *Patterns in Mathematics* – number patterns and series
- *Perimeter and Area*
- *Prime Time* – primes, factors, HCF and LCM
- *Fractions*

Class 7 · Ganita Prakash

- *Working with Fractions* – fractions
- *Finding Common Ground* – primes, factors, HCF and LCM
- *Finding the Unknown* – simple equations

Class 7 · Curiosity (science)

- *Measurement of Time and Motion* – speed, distance and time

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems

Also asked, though the newer NCERT books teach it in a later class: averages (mean)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Curiosity

- *Exploring Magnets* – magnets
- *A Journey through States of Water* – evaporation and condensation

Class 7 · Curiosity

- *Exploring Substances: Acidic, Basic and Neutral* – acids, bases and indicators
- *Electricity: Circuits and their Components* – electric circuits
- *Changes around Us: Physical and Chemical* – physical and chemical changes, rusting
- *Heat Transfer in Nature* – heat and conductors
- *Measurement of Time and Motion* – speed
- *Life Processes in Animals* – respiration
- *Life Processes in Plants* – photosynthesis and plant respiration
- *Light: Shadows and Reflections* – light, shadows and plane mirrors

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS

   Find the next number in the series: 7, 10, 16, 25, 37, ___

   A 49 **B** 52

   C 55 **D** 61

2. ENGLISH

   The old bridge was too fragile to carry the loaded truck. As used here, "fragile" most nearly means

   A ancient **B** narrow

   C weak **D** crowded

3. SCIENCE

   Which of these will a magnet attract?

   A a copper wire **B** an iron pin

   C aluminium foil **D** a glass bead

4. MATHS

   What is the largest number that divides both 48 and 72 exactly?

   A 24 **B** 36

   C 12 **D** 6

5. ENGLISH

   Choose the correct word: Each of the players ___ given a new jersey.

   A were **B** was

   C are **D** have

6. SCIENCE

   Which of these tastes sour because it contains an acid?

   A vinegar **B** soap solution

   C baking soda solution **D** chalk powder

7. MATHS

   A square garden has a perimeter of 48 m. What is its area, in square metres?

   Type-in answer: _____

8. ENGLISH

   Pen is to writer as brush is to ___

   A painter **B** canvas

   C colour **D** wall

9. SCIENCE

   A ball rolls 45 m across flat ground in 9 seconds at a steady speed. What is its speed, in metres per second?

   Type-in answer: _____

10. MATHS

Riya cycles to school at 12 km/h and reaches in 25 minutes. How far is her school from home?

A 3 km
B 4 km
C 5 km
D 6 km

11. ENGLISH

What comes next in the series? AZ, BY, CX, DW, ___

A DV
B EU
C FX
D EV

12. SCIENCE

Two identical bulbs are connected in parallel to a battery and both glow. If one bulb is unscrewed, the other bulb

A goes out
B glows dimmer
C stays lit as before
D flickers and fails

13. MATHS

Asha spends one third of her pocket money on books and one quarter of what is left on snacks. She has ₹150 left. How much pocket money did she have?

A ₹200
B ₹250
C ₹300
D ₹360

14. ENGLISH

Some pilots are engineers. All engineers are graduates. Which statement must be true?

A All pilots are graduates
B No pilot is a graduate
C Some pilots are graduates
D All graduates are pilots

15. SCIENCE

A leaf on a plant is partly covered with black paper for two days, then tested with iodine. Only the uncovered part turns blue-black. This shows that

A iodine reacts with the black paper
B the covered part lost all its water
C the leaf breathes only where it is uncovered
D starch is made only where light reaches

Answers

1	**B. 52**	The gaps grow by 3 each time: +3, +6, +9, +12, so the next gap is +15, and 37 + 15 = 52.
2	**C. weak**	Fragile means easily broken. The sentence says the bridge could not carry the weight, which points to weakness, not age or width.
3	**B. an iron pin**	Magnets attract iron and a few other metals. Copper and aluminium are metals but are not attracted, and glass is not a metal.
4	**A. 24**	48 = 24 × 2 and 72 = 24 × 3, and 2 and 3 share no common factor, so 24 is the largest. 36 does not divide 48.
5	**B. was**	The subject is "each", which is singular, so the verb is "was". "Players" sits nearby but is not the subject.
6	**A. vinegar**	Vinegar contains acetic acid, which gives it a sour taste. Soap and baking soda are bases, and chalk is neither sour nor acidic.
7	**144**	A square has four equal sides, so each side is 48 ÷ 4 = 12 m, and the area is 12 × 12 = 144 square metres. This is a type-in question.
8	**A. painter**	A pen is the tool a writer works with; a brush is the tool a painter works with. The other options are things a brush touches.
9	**5**	Speed is distance divided by time: 45 m ÷ 9 s = 5 m/s. This is a type-in question: the answer is typed as a number.
10	**C. 5 km**	At 12 km/h she covers 1 km every 5 minutes, so in 25 minutes she covers 5 km. The minutes have to be turned into hours first.
11	**D. EV**	The first letter moves forward one step (A, B, C, D, E) while the second moves back one step (Z, Y, X, W, V), giving EV.
12	**C. stays lit as before**	In a parallel circuit each bulb has its own path to the battery, so removing one leaves the other's path complete and unchanged.
13	**C. ₹300**	After books, 2/3 is left; snacks take a quarter of that, leaving 3/4 of 2/3, which is 1/2. Half is ₹150, so she had ₹300. ₹360 comes from taking a quarter of the whole.
14	**C. Some pilots are graduates**	The pilots who are engineers must be graduates, so at least some pilots are graduates. Nothing is said about the other pilots.
15	**D. starch is made only where light reaches**	Iodine turns blue-black where there is starch. Only the part that received light made starch, so the leaf needs light to make food.